

CE 311: Geotechnical Engineering I

Course Code	CE 311
Course Name	Geotechnical Engineering I
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG students (B.Tech. 3rd Year)
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)
Recorded from	CE 302 by the 52nd BoA (Nov 2023)

Course Contents

- **Introduction:** Origin, Formation, Importance of geotechnical engineering, Idealization of soil matrix, three phase diagram, Mass-weight-volume relationships, applications (8 Lectures)
- **Properties of soil:** Particle size distribution, Index properties, Atterberg Limits, plasticity chart, Soil Classification (4 Lectures)
- **Soil Compaction:** Soil density-water relationship, Proctor tests, field tests and applicability (3 Lectures)
- **Permeability in soils:** One-dimensional flow, Darcy's Law, Laboratory and field method of determination of permeability. concept of seepage (6 Lectures)
- **Stresses in soils:** Stresses in soil mass, capillarity phenomena, concept of effective stress, Stresses from elastic theory, Boussinesq and Westergaard's theory, pressure distribution diagram, pressure bulb, shapes of loading area (7 Lectures)
- **Consolidation:** ID laboratory test, preconsolidation pressure, concept of normally consolidated and over consolidated soil, One-dimensional equation and solution, determination of consolidation settlement (5 Lectures)
- **Shear Strength of Soil:** Mohr circle, Mohr-coulomb failure criteria, shear strength of sand and clays, pore pressure, factors affecting shear strength, laboratory measurement of shear strength. (8 Lectures)

Text Books

1. Robert D Holtz, William D. Kovacs and Thomas C. Sheahan, **Introduction to Geotechnical Engineering**, 2nd Edition, Pearson, 2021.
2. V. N. S. Murty, **Geotechnical Engineering**, Book World Enterprises, 2005.

References

1. R.F. Craig, **Soil Mechanics**, ELBS and Van Nostrand Reinhold Co. Ltd., 2004.
2. Braja M. Das, **Principles of Geotechnical Engineering**, 7th Edition and above, Cengage Learning.
3. G. Ranjan and A.S.R. Rao, **Basic and Applied Soil Mechanics**, New Age International Publishers, 2007.